Ex.no:12 Discovery of live host using
20/9/26 ARP, ICMP, TCP, IUDP.

AIM:

To discover live host in a network using Nmap scans such as ARP, ICMP, TCP & IUDP on the TryHackme platform.

OBJECTIVE:

To understand how map identifies active systems before performing further network scanning.

SOFTWARE:

Nmap.

Tryhack Nmap learn

computer with network success.

THEORY:

Nmap is a free and open-source network scanning tool used to discover live hosts and identify network services. It was created.

Nmap uses different live hosts using different protocols.

1) ARP scan - Uses ARP scans request to find hosts on same subnet.

2) ICMP scan - Use ICMP request and replies to identify active host.

3) TCP scan - Use TCP requests and packets to specified ports & check the response.

4) UDP scan - send UDP packets & can identify an active host through an ICMP port.

OBSERVATION:

→ ARP request can discover hosts within the same subset.

→ ARP cannot cross routers because it operates at the link layer.

→ An ARP ping generally requires a previledged account.

→ UDP ping can identify an active host when an ICMP port unreachable is received.

→ The -sn option allows Nmap to perform host discovery without conducting a port scan.

RESULT:

Thus, the live host discovery using Nmap was successfully studied and performed. The experiment helped in understanding how active hosts can be identified before performing further network scans.